

Smooth Everywhere, Universal on the Integers

Canonical quartic certificates with five extra variables,
no local obstruction, and explicit Jacobian identities

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Developed with ChatGPT

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Abstract

We give a uniform arithmetic-geometric finalizer for Diophantine computation. Given any finite system of quadratic integer equations in n variables, it constructs one polynomial of exact degree four in $n + 5$ variables. Its natural zeros are in explicit bijection with the original natural zeros, and its integer zeros are exactly two sign-related copies of the original integer zeros. Nevertheless, its affine scheme is smooth over $\mathbb{Z}$, every geometric fibre is integral, a nonnegative dyadic rational point can always be constructed, and there are integral p -adic points for every prime. Rational points are dense in the real locus, which has exactly two path components. A degree-bounded integral Bézout identity certifies smoothness without a genericity search.

The construction yields a sharply specified undecidable class of smooth, everywhere locally soluble quartic equations. It also preserves canonical bounded-execution certificates without increasing witness multiplicity. We supply a counter-machine frontend, a complete worked export, a sharp weighted-guard obstruction in this architecture, and 21,128 successful symbolic and finite checks. General undecidability for smooth varieties is known; the contribution proposed here is the explicit normal form and its combined guarantees, not a claim of priority for that general phenomenon.

Status. Ordinary mathematical proofs and executable exact checks. Not a Lean/Coq formalization; not peer reviewed. No finite-fold MRDP theorem, no solution to Hilbert’s tenth problem over $\mathbb{Q}$, and no improvement to the repository’s universal operation-count records is claimed.

[write] Written into ProveIt’s research-report collection (*hilbert-tenth-problem*) from batch-79 manuscript 13. “Smooth” means the affine model is smooth over $\mathbb{Z}$; Lemma 2.2 and Theorem 10.1 re-present results of the collection’s report *canonical-diophantine-certificates*. See the [editorial preface](#) and Appendix D.

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Editorial preface

[write] This preface, the bracketed notes marked [write] in the text, Appendix D and the bibliography entries marked [write] were written when the manuscript was placed in ProveIt’s research-report collection. Everything else is the manuscript as delivered, with its numbering of sections, statements and equations. Its labels gained the prefix `sdf:`, and labels were added to three numbered results that had none (Theorem 9.1, Corollaries 9.4 and 11.2), to some sections, to the research questions and to Appendix B.

The source. The report is built from one manuscript, batch-79 manuscript 13 of ProveIt’s incoming reports (archive `Smooth_Quartic_Diophantine_Certificates.zip`, arrival commit `ef2fc7990`, placed by commit `224ca41df`). It names the repository commit `a845ab5d0` as the snapshot it compared against (Section 1.1 and Appendix D). It is an AI-assisted research manuscript (“Developed with ChatGPT”) prepared for Vladimir Reshetnikov. Its proofs are conventional and its checks are finite exact computations; it is unrefereed, and nothing in it is formalized in Lean or Rocq.

Already reviewed. After placement, the Hilbert’s-tenth research tree reviewed the archive (commit `fd4a2e8d3`; files `review_smooth_quartic_aebfa386e.md`, `.py` and `.json` in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`, indexed in `incoming_substrate_review_aebfa386e.md`) [9]. Its verdict is “PASS within the stated mathematical and implementation scope”: it replayed all 21,128 assertions of the delivered verifier and reproduced both example exports byte for byte, checked the integral Jacobian identity in every characteristic, the fibre bijections, geometric integrality, the dyadic and 2-adic constructions and the restricted guard-weight optimality, and ran independent checks (12,636 integer and 640 natural fibre tuples, 120 counter runs against a separate interpreter). It found no theorem defect and no required repair. It stresses five scope points that the text below already states: the size bounds of Proposition 3.2 count monomials, not arithmetic operations; “degree four” is witness degree, and a loader parameter treated as a coordinate gives total degree six (Section 11.3); the counter frontend has an external horizon and accepts halting *by* that horizon, so it is not a fixed-arity compiler; fixed-arity universality comes only from an existing MRDP relation and proves no single-fold MRDP theorem; and the code is not an authenticated checker for edited certificate objects or hostile JSON. The finalizer gives no smaller operation schedule for the repository’s 87-operation universal polynomial.

What is new and what is re-proved. The main new content is the arithmetic geometry of the final equation: smoothness of the affine model over $\mathbb{Z}$ with an explicit integral Bézout certificate (Sections 5–6), local and dyadic points (Section 7), rational density and the two real components (Section 8), the weighted-guard classification (Section 9) and the power-of-two padding (Section 12). Three pieces re-present results already in the collection’s report *canonical-diophantine-certificates* (CDC below) [7]; they are printed because the argument uses them, with [write] notes at each and no novelty claimed:

- Lemma 2.2, natural version, is CDC’s unique-extension quadratization, `cdc:wf:lem:quadratization`, with the same proof ($F^+ - F^-$ circuits, one natural variable per gate, uniqueness by induction along the circuit); the integer version with subtraction gates is the same induction. The sum-of-squares step $H = \sum q_i^2$ is CDC’s `cdc:wf:eq:sos`.
- Theorem 10.1 is a second presentation of CDC’s counter-program adapter: one-hot control equations, the zero guard (25), which is CDC’s `cdc:eq:zero-guard`, the externally

fixed horizon of CDC’s Section `cdc:pt:sec:boundary`, and the halt self-loop caveat of `cdc:ct:sec:substrates`. New here are only the exact counts n_T , m_T and the positive-edge slack equation (26) with natural selectors summing to one in place of Boolean rows.

- The computability core of Theorem 11.1 is CDC’s quartic normalization, `cdc:of:prop:quartic` (every computably enumerable set has a quartic representation by MRDP, circuit lifting and a sum of squares), together with Poonen’s smooth-variety undecidability, which the manuscript credits itself. The new content of that theorem is the geometric package (a)–(e).

Three further credits are added as [write] notes: Ramanujan’s list of universal diagonal quaternary forms for Lemma 7.1, Minsky’s two-counter universality beside the formalized reductions of Section 10, and a variable count beside Poonen’s lemma in Section 1.2.

A non-claim of the collection met in finalized form. The report *probabilistic-quantum-and-continuous-computation* (PQC) [8] proves that each zero of its quartic Boolean-circuit energy $\mathcal{E}_C$ is a nondegenerate minimum (`pqc:ql:thm:hessian`) and, in the remark “Non-degenerate minimum versus smooth hypersurface” that follows it, does not claim that the hypersurface $\mathcal{E}_C = 0$ is nonsingular. It cannot be: $\mathcal{E}_C$ is a sum of squares, so its gradient vanishes at every real zero, which is the observation that opens Section 1 below. The residuals of $\mathcal{E}_C$ are integer polynomials of degree at most two, so Theorem 3.1 applies to them. Five further variables give a quartic whose affine hypersurface is smooth over $\mathbb{Z}$, with the same natural (here Boolean) zeros. The price is positivity. The finalized polynomial takes negative real values, so its zeros are no longer energy minima, and the Morse statement of PQC and the smoothness statement here concern different polynomials. Neither report proves the other’s statement.

Relation to CDC Part XI. CDC’s Part XI classifies representations by polynomials that are nonnegative on the whole real orthant (the classes $\mathcal{D}_d^+$ of `cdc:of:thm:classification`). The finalized polynomial F of (3) is never in such a class: at the dyadic point (13) the coordinates t, u, z_j are positive, and $F_u = 2tR = 2B/K^3$ and $F_{z_j} = \lambda_j(4z_j - 1) = 2\lambda_j A_j/K^2$ are not all zero, because $A_1^2 + A_2^2 + 2A_3^2 + 2B^2 = K^4 - 2c > 0$. Decreasing one coordinate with a nonzero derivative slightly gives a point with nonnegative coordinates where $F < 0$. Some sign change is forced: a polynomial that is nonnegative near one of its zeros has zero gradient there, so a hypersurface smooth at a zero inside the open orthant changes sign there. The manuscript uses positivity only on the integer lattice (Lemma 4.1), which is among the settings that CDC’s README, in its scope note for manuscript 09 (CDC’s Part XI), says that Part’s theorem does not cover (“positivity only on natural points”).

Batch-79 siblings. Several manuscripts of the same batch end with an ordinary sum of squares of integer residuals of degree at most two in natural unknowns, which is exactly the input of Theorem 3.1: manuscript 03’s bounded-bit quartic (placed for CDC’s Part XVII), and the bounded ordinary exports of manuscripts 06 and 15 (“Coefficient extraction at an external bound” and “Bounded ordinary quartic export”, Parts I and II of the report *polynomial-witness-histories*). Each could be followed by this finalizer at the cost of five variables; no source states that composition, and none cites this manuscript, which arrived after four of them. Manuscript 02’s exports keep fixed-base power atoms and are not ordinary polynomial equations, so the finalizer does not apply to them as they stand; the unbounded exports of 06 and 15 have polynomial-valued unknowns and are outside its integer setting. Manuscript 15’s

question whether one equation can have degree two or three is not answered here: the finalizer stays at degree four.

Notation. The manuscript reuses several letters between its algebraic sections and its counter frontend, and some coincide with letters of neighbouring reports. No symbol was renamed. The table lists the readings; the right column gives the other reading in this report, or the tempting false one.

Symbol or term	Main meaning	Other meaning in this report, or the tempting false reading
F	the finalized quartic (3) in $n + 5$ variables	not nonnegative over $\mathbb{R}$ or on the orthant; not CDC's MRDP polynomial $F(x, y)$ nor PQC's energy $\mathcal{E}_C$
g	the guard $g(z) = 2z^2 - z$	Section 10: g is the number of guarded edges
E	the Euler combination $E = 4H$ ((6), Section 9)	Section 10: the number of edges
D_j	$D_j = F_{z_j}$ (Section 5)	countdown example: D is the decrement edge
H	$\sum_i q_i^2$	countdown example: the halt self-loop edge
t, q_i, c	homogenizing coordinate; homogeneous lift; $c = \sum_i f_i(0)^2$ (11)	Section 10: target state $t(e)$, state variable $q_{s,i}$, counter $c_{s,j}$
p, d, h	a prime; the power-of-two degree of Theorem 12.1; $h(z_1)$ of (14)	Section 10: number of positive-decrement edges, slack $d_{s,e}$, halt state h
B	the coordinate with $K^4 - 2c = A_1^2 + A_2^2 + 2A_3^2 + 2B^2$ (Theorem 7.2)	Corollary 4.2: the box radius
w_j, W	guard weights and their total (Section 9)	w : a gate variable (Lemma 2.2) and the second guard variable of (21); w_j : rescaled guard coordinates in the proof of Theorem 8.2
N	the integer $K^4 - 2c$ (Section 7)	not $\mathbb{N} = \{0, 1, 2, \dots\}$ (as in CDC)
smooth, good reduction	the affine scheme $\mathcal{X} \rightarrow \text{Spec } \mathbb{Z}$ is smooth	not smoothness of a projective closure, nor a smooth proper model (neither claimed)
dyadic point	a point of $\mathcal{X}(\mathbb{Z}[1/2])$, here with nonnegative coordinates	not density of dyadic points (not claimed)
degree four	degree in the witness variables of a specialized equation	not total degree with input parameters as coordinates (six, Section 11.3)
canonical, witness-faithful finalizer	natural zeros in bijection with the source's a post-processing of a quadratic system into one equation	not single-fold or finite-fold MRDP (not claimed) not the operation-count “finalizers” of the Hilbert's-tenth research tree; this one costs more operations than a sum of squares

1 The problem: preserve computation, simplify the geometry

A computational certificate normally begins as a list of equations. States, registers, scheduling choices, memory contents, and auxiliary multiplication gates are constrained by polynomial residuals. Combining those residuals by squaring is convenient over the integers, but it destroys geometric regularity: if

$$P(x) = \sum_{i=1}^m f_i(x)^2,$$

then every common zero of the f_i is a critical point of P . This is not an incidental implementation defect. The ordinary sum-of-squares finalizer makes precisely the computationally meaningful points singular.

The question considered here is more demanding than whether computation has some Diophantine representation:

Can a certificate be replaced by one low-degree equation, without changing its natural-number witnesses, while making the affine model smooth at every prime and making all local and rational existence tests automatically succeed?

The answer is affirmative. The resulting separation is between lattice arithmetic and geometry: smoothness, rational points, real connectedness, and local integral points can all be harmless while integer solvability still carries an arbitrary computably enumerable problem.

1.1 Relation to the repository

The ProveIt repository already has a substantial collection of witness-faithful Diophantine frontends. The reviewed canonical-certificate collection includes bounded causal traces, queue and memory histories, dynamic heaps, interaction nets, and stabilization certificates. Its status document explicitly separates ordinary proofs, finite Python checks, and unproved uniform finite-fold claims [4, 5, 6]. This manuscript adds a different layer: a *finalizer* that can follow any quadratic frontend, irrespective of its underlying substrate.

The distinction matters. Replacing a scheduler, changing a memory encoding, or quotienting commuting histories is a semantic task. Our finalizer neither performs nor assumes such a quotient. It takes whatever witness set has already been justified and preserves it exactly over $\mathbb{N}$. Thus a unique certificate remains unique, a family indexed by trace classes retains those classes, and an existing noncanonical representation remains noncanonical.

The targeted repository comparison used the canonical-certificate status file at commit

a845ab5d0f4591294ef3c2d92feab5825a8d543.

Other overview files were read on the live branch. This was not an exhaustive repository audit, a build, or a verification of every theorem mentioned in its documentation.

[write] [At the write, the status file read at this commit is unchanged (same blob, Appendix D). Two of the frontend tools used below are already in the collection’s report *canonical-diophantine-certificates*, and a non-claim of the report *probabilistic-quantum-and-continuous-computation* bears on the main theorem; the [editorial preface](#) gives the pointers.]

1.2 Known antecedent and the precise novelty boundary

Poonen proved that integer solvability can be preserved while replacing an equation by one defining a smooth geometrically integral affine variety over $\mathbb{Q}$. His Lemma 4.1 uses a

positive multiple of $y^2 - y$ added to a square, with a suitable coefficient chosen effectively. He also obtained undecidability under a smooth-projective-closure condition [1, Lemma 4.1 and Corollary 4.3].

Consequently, *smoothness does not make Hilbert's tenth problem decidable* is not a new claim. The proposed contribution here is the simultaneous, coefficient-independent package: five additional variables, a quartic after quadratic lifting, an exact natural-fibre bijection, smoothness over the entire integer base, an integral Jacobian certificate of bounded degree, and explicit dyadic and local points. A comprehensive priority search for this exact combination was not completed. The construction is presented as a research result with proofs, not as a certified world-first theorem.

[write] [Variable count: Poonen's lemma adds one variable to a single equation f and gives degree $2 \deg f$ and smoothness over $\mathbb{Q}$; applied to $f = \sum_i f_i^2$ for a quadratic system it would give degree eight. The price here of smoothness over all of $\mathbb{Z}$, with an explicit certificate, degree four and a natural-zero bijection, is four further variables. The Hilbert's-tenth review [9] also read Poonen's Lemma 4.1 and Corollary 4.3 and found that they support this attribution and do not contain the five-variable formula; it was not a complete priority search.]

1.3 Reading map

Sections 3–6 prove the algebraic finalizer. Sections 7–8 prove the arithmetic and real-geometric claims. Section 9 explains why three weighted guards are the smallest successful choice in the specified weight-four architecture. Sections 10–11 connect the finalizer to computation, carefully separating external time bounds from fixed-arity unbounded representations. The implementation and its limits are recorded in Section 13.

2 Conventions and the quadratic interface

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. A polynomial with integer coefficients is evaluated in $\mathbb{Z}$ even when its variables range over $\mathbb{N}$. Negative coefficients in a residual do not change the domain of its witnesses. For a finite system $\mathcal{S} = (f_1, \dots, f_m)$, put

$$Z_D(\mathcal{S}) = \{a \in D^n : f_i(a) = 0 \text{ for every } i\}, \quad D \in \{\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}\}.$$

Empty systems and zero residuals are allowed. Parameters are coefficients unless explicitly declared to be additional variables.

Definition 2.1 (Quadratic interface). A quadratic certificate consists of named variables $x = (x_1, \dots, x_n)$, a finite list of polynomials $f_i \in \mathbb{Z}[x]$ of total degree at most two, and a separately justified interpretation of its zeros over a stated domain. The interpretation is not part of the algebraic data.

2.1 Functional arithmetic-circuit lifting

Lemma 2.2 (Quadratic lifting). *A finite system of integer polynomial equations can effectively be replaced by a quadratic system whose projection onto the original variables is a bijection of integer zero sets. Over $\mathbb{N}$, the same statement holds using nonnegative arithmetic circuits for the positive and negative parts of each polynomial.*

Proof. Over $\mathbb{Z}$, evaluate each polynomial by a finite straight-line circuit. For an addition,

subtraction, or multiplication gate introduce a fresh variable w and the corresponding equation

$$w - a - b = 0, \quad w - a + b = 0, \quad w - ab = 0.$$

Constants are imposed by $w - c = 0$, and output variables are set to zero. Every source tuple has exactly one gate assignment, by induction on the acyclic circuit order. The residuals have degree at most two, and the outputs vanish exactly when the original equations hold.

Over $\mathbb{N}$, first write $f = f^+ - f^-$ with nonnegative integer coefficients. Evaluate f^+ and f^- using addition and multiplication only, and equate the two output variables. All gate values are then natural, and the same induction proves uniqueness. A circuit using arbitrary subtraction is not silently treated as a natural circuit. $\square$

This lemma preserves *gate* uniqueness. It does not make the witnesses of a pre-existing Diophantine representation unique. For example, a representation of halting may have many auxiliary solutions before any circuit gates are introduced. That multiplicity survives lifting.

[write] [The natural version of Lemma 2.2 is the unique-extension quadratization of the collection's report *canonical-diophantine-certificates*, Lemma `cdc:wf:lem:quadratization`, with the same proof: circuits for f^+ and f^- , one natural variable per gate, uniqueness by induction along the circuit. The integer version adds subtraction gates to the same induction. No novelty is claimed for either; the lemma is printed because Theorem 11.1 uses both versions.]

2.2 Affine smoothness and degree conventions

Our geometric object is the affine scheme

$$\mathcal{X} = \text{Spec}(\mathbb{Z}[x, t, u, z_1, z_2, z_3]/(F)).$$

Smoothness over $\mathbb{Z}$ means that its structure morphism to $\text{Spec } \mathbb{Z}$ is smooth, not merely that its real locus has no singularities. A geometric fibre is obtained by passing to an algebraic closure of a residue field of $\mathbb{Z}$, including the generic characteristic-zero fibre.

We use the standard hypersurface criterion: if the ideal generated by F and all its partial derivatives is the whole polynomial ring, the derivative-open subsets of the hypersurface are standard smooth and cover it [3]. In particular, an integral identity expressing 1 in that ideal certifies relative smoothness in every characteristic.

Degree four will always mean degree in the *witness variables of a specialized equation*. When input parameters become coordinates of a single master polynomial, their contribution to total degree must be counted again; Section 11.3 does so explicitly.

3 The five-variable finalizer

Write each source residual as

$$f_i(x) = \sum_{|\alpha| \leq 2} c_{i,\alpha} x^\alpha.$$

Its homogeneous quadratic lift is

$$q_i(x, t) = \sum_{|\alpha| \leq 2} c_{i,\alpha} x^\alpha t^{2-|\alpha|}. \tag{1}$$

The expression $t^2 f_i(x/t)$ is a useful mnemonic for $t \neq 0$; the polynomial definition (1), valid also at $t = 0$, is authoritative. Set

$$H(x, t) = \sum_{i=1}^m q_i(x, t)^2, \quad R = tu - 1, \quad g(z) = 2z^2 - z. \quad (2)$$

Put $\lambda = (\lambda_1, \lambda_2, \lambda_3) = (1, 1, 2)$. The finalizer is

$$F(x, t, u, z) = H(x, t) + (tu - 1)^2 + g(z_1) + g(z_2) + 2g(z_3). \quad (3)$$

Here $z = (z_1, z_2, z_3)$, and all five new variables are distinct from the source variables.

Theorem 3.1 (Canonical smooth quartic finalizer). *For every quadratic interface in n variables, the polynomial (3) has the following properties.*

- (i) *Its coefficients are integers and its total degree is exactly four.*
- (ii) *The map $a \mapsto (a, 1, 1, 0, 0, 0)$ is a bijection from $Z_{\mathbb{N}}(\mathcal{S})$ to $Z_{\mathbb{N}}(F)$.*
- (iii) *The map*

$$(a, \varepsilon) \mapsto (\varepsilon a, \varepsilon, \varepsilon, 0, 0, 0)$$
is a bijection from $Z_{\mathbb{Z}}(\mathcal{S}) \times \{1, -1\}$ to $Z_{\mathbb{Z}}(F)$.
- (iv) *The affine scheme $\mathcal{X}$ is smooth over $\mathbb{Z}$ of relative dimension $n + 4$, and every geometric fibre is integral.*
- (v) *An explicit identity $AF + \sum_v B_v F_v = 1$ exists with $\deg A \leq 3$ and $\deg B_v \leq 4$; its coefficients are given below without an elimination or genericity search.*
- (vi) *A point of $\mathcal{X}(\mathbb{Z}[1/2])$ with nonnegative rational coordinates can always be constructed. Also $\mathcal{X}(\mathbb{Z}_p) \neq \emptyset$ for every prime p , and $\mathcal{X}(\mathbb{Z}/M\mathbb{Z}) \neq \emptyset$ for every $M \geq 1$.*
- (vii) *$\mathcal{X}(\mathbb{Q})$ is dense in $\mathcal{X}(\mathbb{R})$ and is Zariski dense in $\mathcal{X}_{\mathbb{Q}}$. The real locus has exactly two path components.*

The theorem is proved in the next five sections. Notice how different its claims are: the first three are statements about an integer lattice, the fourth and fifth about schemes, and the last two about rational, real, and local points. None is being substituted for another.

Proposition 3.2 (Size bounds). *If f_i has s_i nonzero monomials, the number of monomials of F is at most*

$$9 + \sum_{i=1}^m \binom{s_i + 1}{2}.$$

For the sum of absolute values of coefficients,

$$\|F\|_1 \leq \sum_i \|f_i\|_1^2 + 16.$$

The construction of the expanded polynomial has polynomial bit complexity in the expanded quadratic input size.

Proof. Homogenization preserves the number and coefficients of source monomials. A square of an s_i -term polynomial contributes at most $s_i(s_i + 1)/2$ monomials. The unit square contributes at most three and the three guards contribute six. Possible coincidences only lower the count. The ℓ^1 norm is submultiplicative; the unit square has norm four and the three weighted guards together have norm twelve. The stated expansion uses a quadratic number of input-term products with polynomial-length coefficients. $\square$

These are expanded-polynomial bounds, not claims about the smallest arithmetic circuit or the repository's universal operation-count records.

4 Exact integer fibres and witness heights

Lemma 4.1 (A one-zero integer guard). *For every $z \in \mathbb{Z}$, $g(z) \geq 0$, with equality if and only if $z = 0$. For every $z \in \mathbb{R}$,*

$$g(z) = 2 \left(z - \frac{1}{4} \right)^2 - \frac{1}{8}. \quad (4)$$

Proof. For $z \geq 1$, $z(2z - 1) > 0$; for $z \leq -1$, both factors are negative, so again their product is positive. The remaining integer is zero. Completing the square gives (4). $\square$

Proof of Theorem 3.1(i)–(iii). Each q_i is homogeneous of degree two, so H has degree at most four. The monomial t^2u^2 has coefficient one in F : no term of H involves u , and the guards involve neither t nor u . Thus the degree is exactly four, including for the empty or identically zero source system.

On an integer tuple, every summand in (3) is nonnegative. Consequently $F = 0$ forces

$$q_i(x, t) = 0 \quad (1 \leq i \leq m), \quad tu = 1, \quad z_1 = z_2 = z_3 = 0.$$

The units of $\mathbb{Z}$ are ± 1 , so $t = u = \varepsilon$ for one $\varepsilon \in \{1, -1\}$. Put $a = \varepsilon x$. Homogeneity gives

$$q_i(\varepsilon a, \varepsilon) = \varepsilon^2 f_i(a) = f_i(a).$$

Thus a is a source zero. Conversely, every displayed lift of a source zero makes every summand vanish. The inverse map is $(x, t, u, z) \mapsto (tx, t)$, proving bijectivity and not just existence.

If all coordinates are natural, $tu = 1$ forces $t = u = 1$. The resulting inverse map is simply projection onto x . No sign choice remains. $\square$

Corollary 4.2 (Exact counting and height transfer). *Let $B \geq 1$ be an integer, and count zeros whose coordinates have absolute value at most B . The number of target natural zeros is exactly the number of source natural zeros; the number of target integer zeros is twice the number of source integer zeros. For each individual lift,*

$$\text{ht}(\varepsilon a, \varepsilon, \varepsilon, 0^3) = \max\{1, \text{ht}(a)\}, \quad \text{ht}(a) = \max_i |a_i|.$$

The same assertions hold fibrewise after any external parameters are fixed.

Proof. The bijections just proved add only ± 1 and zero and change source coordinates only by a common sign. They therefore preserve every box of radius $B \geq 1$ in the asserted manner. $\square$

Remark 4.3 (What the extra coordinates do not encode). The three z coordinates are zero at every integer solution. They are not additional computational choices. The t, u coordinates impose a unit condition, and over $\mathbb{N}$ that unit is uniquely one. Their geometric role becomes visible only away from the integer lattice, where the guards may be negative and t may be a nonintegral unit.

5 Smoothness in every characteristic

There are two complementary proofs. The first explains the choice of constants through critical values. The second gives an integral identity suitable for independent symbolic checking and eventual formalization.

5.1 The critical-value mechanism

Put $Q = H + R^2$, so $F = Q + \sum_j \lambda_j g(z_j)$.

Lemma 5.1 (Only two possible critical values). *Over any field of characteristic different from two, every critical point of Q has value either zero or one.*

Proof. Homogeneity of H gives the polynomial identity

$$\sum_{i=1}^n x_i Q_{x_i} + t Q_t - u Q_u = 4H. \quad (5)$$

The contributions of R^2 cancel between the last two terms. At a critical point the left side is zero, hence $H = 0$. Moreover $Q_u = 2tR$. If $t \neq 0$, then $R = 0$ and $Q = 0$. If $t = 0$, then $R = -1$ and $Q = 1$. $\square$

A singular point of $F = 0$ in odd characteristic would satisfy $4z_j - 1 = 0$ for all three guards. Their weighted values sum to $-4/8$, so $Q = 1/2$. That contradicts Lemma 5.1. In characteristic two, $F_{z_1} = 4z_1 - 1 = 1$, which rules out singularities directly.

This calculation is coefficient independent. In particular, no assumption about the source zero set, its dimension, or the regularity of its defining system is used.

5.2 An integral unit-ideal certificate

For derivatives of the full polynomial F , define

$$\begin{aligned} E &= \sum_{i=1}^n x_i F_{x_i} + t F_t - u F_u = 4H, \\ D_j &= F_{z_j} = \lambda_j(4z_j - 1), \\ J &= 8F - 2E - \sum_{j=1}^3 (4z_j - 1)D_j = 8R^2 - 4. \end{aligned} \quad (6)$$

The rightmost equalities are identities over $\mathbb{Z}$, not reductions on the zero set.

Proposition 5.2 (Universal Jacobian identity). *In $\mathbb{Z}[x, t, u, z]$,*

$$4u(2R - 1)F_u - (2R + 1)J = 4 \quad (7)$$

and therefore

$$\boxed{z_1[4u(2R - 1)F_u - (2R + 1)J] - D_1 = 1.} \quad (8)$$

In particular, 1 belongs to the ideal generated by F and its partial derivatives.

Proof. Substitute $F_u = 2tR$ and $J = 8R^2 - 4$ into the left side of (7), and use $tu = R + 1$. It becomes

$$8R(R + 1)(2R - 1) - (2R + 1)(8R^2 - 4) = 4.$$

Multiplying by z_1 and subtracting $D_1 = 4z_1 - 1$ gives (8). By (6), both E and J are explicit combinations of the equation and its derivatives. Every product $(4z_j - 1)D_j$ is a derivative times a polynomial, so it also belongs to the same ideal. $\square$

For direct certificate exchange, no nested abbreviations are necessary. Let $F_v = \partial F / \partial v$, and set

$$\begin{aligned} A &= -8z_1(2R+1), \\ B_{x_i} &= 2z_1(2R+1)x_i, \quad B_t = 2z_1(2R+1)t, \\ B_u &= 2z_1u(2R-3), \quad B_{z_j} = z_1(2R+1)(4z_j-1) - \delta_{j1}. \end{aligned} \tag{9}$$

Expansion of (8) gives

$$\boxed{AF + \sum_v B_v F_v = 1.} \tag{10}$$

Here $\deg A \leq 3$ and $\deg B_v \leq 4$, so every summand in this certificate has degree at most seven. The degree bound is independent of the number of source equations and the size of their coefficients.

Proof of smoothness in Theorem 3.1. The identity (10) persists under every base change. On the hypersurface, the partial derivatives generate the unit ideal. The open subsets where one partial is invertible form a standard-smooth cover [3]. Thus $\mathcal{X} \rightarrow \operatorname{Spec} \mathbb{Z}$ is smooth. It is locally one smooth equation in $n+5$ variables, so its relative dimension is $n+4$. This argument does not infer relative smoothness merely from a picture of the real locus or from a finite list of prime checks. $\square$

Remark 5.3 (Universal coefficient families). The same identity works with coefficients in any commutative base ring. In particular, letting the source coefficients vary gives a smooth relative family over the coefficient-parameter scheme. This does not assert that there is a polynomially varying dyadic point over that entire parameter space; the point construction below is performed after integer specialization.

6 Geometric integrality and the special fibre at two

Proposition 6.1. *Every geometric fibre of $\mathcal{X} \rightarrow \operatorname{Spec} \mathbb{Z}$ is integral. In characteristic two the fibre is isomorphic to affine $(n+4)$ -space.*

Proof. Let k be an algebraically closed field of odd characteristic or characteristic zero. Regard F as a polynomial in z_1, z_2, z_3 over $K = k(x, t, u)$. Its highest homogeneous part in the z variables is

$$2z_1^2 + 2z_2^2 + 4z_3^2,$$

a quadratic form of rank three. A nontrivial factorization of this degree-two polynomial over K would have two affine-linear factors in z . Their leading product has rank at most two, a contradiction. Hence F is irreducible over K .

As a polynomial in z over $k[x, t, u]$, F is primitive: the coefficient of z_1^2 is the nonzero constant 2. Gauss's lemma now gives irreducibility in $k[x, t, u, z]$, a unique factorization domain. The principal ideal (F) is prime, so the fibre is integral.

In characteristic two, every $2z_j^2$ disappears and the coefficient of z_1 is $-1 = 1$. Solving uniquely for z_1 identifies the coordinate ring with a polynomial ring in the remaining $n+4$ variables. This proves both integrality and the stated affine-space description. $\square$

There is no hidden component carrying the actual computation while another component supplies easy rational points. The entire geometric fibre is one integral variety. The integer and rational behaviours coexist on that single variety.

The phrase “good reduction” in this article refers to the displayed *affine* smooth model. It does not say that the naive projective closure is smooth, or that there is a smooth proper model over $\mathbb{Z}$.

7 Dyadic points and the complete absence of local obstructions

7.1 An explicit point after inverting only two

Let

$$c = \sum_{i=1}^m f_i(0)^2 \in \mathbb{N}. \quad (11)$$

Choose $K = 2^k$ with $K^4 > 2c$, and set $N = K^4 - 2c > 0$. Lagrange’s four-square theorem supplies nonnegative integers $a_1, \dots, a_4$ such that

$$N = a_1^2 + a_2^2 + a_3^2 + a_4^2. \quad (12)$$

We use this classical theorem only for a finite integer, not as a statement about representing computations; it is also the standard bridge from natural to integer witnesses in Diophantine undecidability arguments [1, Section 5].

Lemma 7.1 (An elementary universal weighted form). *Every nonnegative integer is of the form*

$$A_1^2 + A_2^2 + 2A_3^2 + 2B^2, \quad A_1, A_2, A_3, B \in \mathbb{N}.$$

The same form represents every nonnegative rational number over $\mathbb{Q}$.

Proof. Start from four ordinary squares. Among four integers, two have the same parity. Replace the squares of that pair, say a, b , by

$$a^2 + b^2 = 2 \left(\frac{a+b}{2} \right)^2 + 2 \left(\frac{a-b}{2} \right)^2.$$

The other two squares remain unchanged, and an absolute value makes the last coordinate nonnegative. For $A/B \geq 0$, represent the integer AB and divide all four coordinates by B . $\square$

[write] [The form $x^2 + y^2 + 2z^2 + 2w^2$ is one of the universal positive diagonal quaternary forms listed by Ramanujan [10]; the elementary parity proof above derives its universality from Lagrange’s theorem.]

Theorem 7.2 (Constructive dyadic solubility). *Choose a representation*

$$K^4 - 2c = A_1^2 + A_2^2 + 2A_3^2 + 2B^2.$$

Then the coordinates

$$x = 0, \quad t = \frac{1}{K}, \quad u = K + \frac{B}{K}, \quad z_j = \frac{1}{4} + \frac{A_j}{2K^2} \quad (1 \leq j \leq 3) \quad (13)$$

define a point of $\mathcal{X}(\mathbb{Z}[1/2])$, all of whose coordinates are nonnegative. If K is the smallest admissible power of two, the rational coordinate height is $O(\sqrt{c+1})$, and its bit length is $O(\log(c+2))$.

Proof. At these coordinates $R = B/K^2$ and $H = c/K^4$. Completing the weighted squares gives

$$\sum_{j=1}^3 \lambda_j g(z_j) = \frac{A_1^2 + A_2^2 + 2A_3^2}{2K^4} - \frac{1}{2}.$$

Adding R^2 and H yields zero by the displayed representation of $K^4 - 2c$. All denominators are powers of two. The coordinates satisfy $1/4 \leq z_j \leq 3/4$ and $K \leq u \leq 2K$.

For a reduced rational a/b with $b > 0$, use coordinate height $\max\{|a|, b\}$. All coordinates in (13) have height at most $4K^2$. If $c \geq 1$, minimality gives $(K/2)^4 \leq 2c$, hence $K^4 \leq 32c$. The case $c = 0$ has $K = 1$. These inequalities give the claimed bounds. $\square$

The small height bound is not a deterministic polynomial-time claim for finding a four-square decomposition in the bit model. The included reference routine uses a finite pair-table search, and explicitly limits the size of that search. A supplied decomposition can instead be checked immediately by (12).

7.2 Integral points at all primes

Theorem 7.3 (Every integral local test succeeds). *For every prime p , $\mathcal{X}(\mathbb{Z}_p)$ is nonempty. Consequently the equation has a solution modulo every positive integer.*

Proof. For an odd prime p , the dyadic point (13) is already integral in $\mathbb{Z}_p$, because its denominators are powers of two.

For $p = 2$, fix $x = 0$, $t = u = 1$, and $z_2 = z_3 = 0$. The remaining equation is

$$h(z_1) := c + 2z_1^2 - z_1 = 0. \quad (14)$$

Modulo two it has the root $z_1 \equiv c$, and $h'(z_1) = 4z_1 - 1$ is odd. Hensel lifting therefore gives a root in $\mathbb{Z}_2$. To avoid treating the lifting as a black box, suppose $h(z_s) \equiv 0 \pmod{2^s}$ and choose $b \in \{0, 1\}$ so that

$$h(z_s)/2^s + b \equiv 0 \pmod{2}.$$

Then $z_{s+1} = z_s + b2^s$ satisfies $h(z_{s+1}) \equiv 0 \pmod{2^{s+1}}$. The compatible residues converge to the required root.

For $M = \prod p^{e_p}$, reduce one local point modulo each p^{e_p} and apply the Chinese remainder theorem coordinate by coordinate. The resulting point solves $F = 0$ modulo M . $\square$

Thus rejecting source instances cannot be rejected by finding a bad prime, a bad prime power, or any congruence modulus. They even have compatible integral points at every prime. This is a failure of a prospective *integral* local-to-global principle, not a failure of the rational Hasse principle: rational points are already available globally.

7.3 A five-variable counterexample requiring no machine encoding

Example 7.4 (An everywhere soluble equation with no integer point). Take the inconsistent source equation $1 = 0$, with no source variables. Then

$$t^4 + (tu - 1)^2 + g(z_1) + g(z_2) + 2g(z_3) = 0 \quad (15)$$

has no integer solution. Indeed, vanishing forces $tu = 1$ and $t^4 = 0$, which are incompatible. Nevertheless it has the dyadic point

$$(t, u, z_1, z_2, z_3) = \left(\frac{1}{2}, 3, \frac{1}{4}, \frac{1}{2}, \frac{3}{8}\right).$$

Here $K = 2$ and $14 = 0^2 + 2^2 + 2 \cdot 1^2 + 2 \cdot 2^2$. The Jacobian identity proves smoothness over $\mathbb{Z}$, not just at this point; Theorem 7.3 supplies all integral local points.

This example is elementary, but its role in the general reduction is important. The same harmless arithmetic geometry occurs for instances whose integer solvability encodes arbitrary halting behaviour. The phenomenon is not restricted to isolated inconsistent equations that an algorithm could recognize syntactically.

8 Rational density and the real locus

8.1 Rational points in the quadric fibres

At every real point, completing the weighted guard squares gives

$$H + (tu - 1)^2 \leq \frac{1}{2}, \quad tu \geq 1 - \frac{1}{\sqrt{2}} > 0.$$

In particular t is nonzero. We may therefore project onto the base $b = (x, t)$ and recover $u = (R + 1)/t$ from R . Put

$$\rho(b) = \frac{1}{2} - H(x, t), \quad y_j = z_j - \frac{1}{4}. \quad (16)$$

The remaining equation is the ellipsoid

$$R^2 + 2y_1^2 + 2y_2^2 + 4y_3^2 = \rho(b). \quad (17)$$

For rational b with $\rho(b) > 0$, this quadric has a rational point. Indeed, after multiplication by two, its left side becomes

$$(2y_1)^2 + (2y_2)^2 + 2(2y_3)^2 + 2R^2,$$

and Lemma 7.1 represents the positive rational $2\rho(b)$.

A nonsingular ellipsoid with rational coefficients and a rational point has dense rational points in its real locus. A line through the rational point with rational direction meets the quadric in a second rational point: substitution gives a quadratic equation with one known root. The resulting rational parametrization covers the quadric except for the usual lower-dimensional exceptional set, and rational parameters are dense in real parameters. Equivalently, use stereographic projection after fixing a rational point and its rational tangent hyperplane.

Theorem 8.1 (Real and Zariski density). *The set $\mathcal{X}(\mathbb{Q})$ is dense in $\mathcal{X}(\mathbb{R})$ in the Euclidean topology and is Zariski dense in $\mathcal{X}_{\mathbb{Q}}$.*

Proof. Consider a real point with $\rho(b) > 0$. Approximate $b = (x, t)$ by rational base points b_ν with nonzero t_ν and positive ρ . Radially rescale (R, y_1, y_2, y_3) to get nearby real points on the nearby ellipsoids. On each such ellipsoid choose a rational point arbitrarily close to that real point. Recovering $u = (R + 1)/t$ is continuous because $t \neq 0$. This produces rational points tending to the original point.

At a boundary point $\rho = 0$, all four coordinates (R, y) vanish and $H = 1/2$. Euler's identity $\sum x_i H_{x_i} + t H_t = 4H$ shows that ∇H is not zero there. Thus every neighbourhood contains base points with $\rho > 0$. The ellipsoid radii tend to zero on approach to the boundary, so the same approximation also reaches the boundary point.

There is a nonempty real open set with $t \neq 0$ and $\rho > 0$: take $x = 0$ and positive t sufficiently small. The smooth integral variety therefore has real manifold points of its full dimension $n + 4$. A polynomial function vanishing on all its rational points vanishes, by Euclidean density, on this real open manifold. In smooth local algebraic coordinates a regular function vanishing on a real open set vanishes identically on the irreducible variety. Hence the rational points are Zariski dense. $\square$

The last local fact can also be seen by complexifying a smooth coordinate chart: a holomorphic function that vanishes on an open subset of its real coordinate slice has identically zero Taylor series. Geometric integrality then propagates the identity to the whole variety. No density assertion is made for *dyadic* points; ordinary rational directions, with arbitrary denominators, were used in the quadric argument.

8.2 Exactly two path components

Theorem 8.2. *The real locus has exactly two path components, distinguished by the sign of t , or equivalently the sign of u .*

Proof. On $\mathcal{X}(\mathbb{R})$, the sum of the guards is at least $-1/2$. Hence

$$H + (tu - 1)^2 \leq \frac{1}{2}, \quad tu \geq 1 - \frac{1}{\sqrt{2}} > 0.$$

Thus t, u are nonzero and have the same sign. Both signs occur, for example by changing (x, t, u) to $(-x, -t, -u)$ in the point (13). No continuous path can join the two signs.

To show that the positive part is path connected, use coordinates

$$y = x/t, \quad r = tu - 1, \quad w_j = \sqrt{2\lambda_j}(z_j - 1/4), \quad t > 0.$$

If $S(y) = \sum_i f_i(y)^2$, the equation becomes

$$t^4 S(y) + r^2 + \sum_{j=1}^3 w_j^2 = \frac{1}{2}. \quad (18)$$

Conversely, every solution of (18) with $t > 0$ gives an original point: $u = (1 + r)/t > 0$ since $|r| \leq 1/\sqrt{2}$.

For fixed y , decreasing t makes the squared radius $1/2 - t^4 S(y)$ nondecreasing. A point can therefore be moved to arbitrarily small positive t by radially rescaling its vector (r, w) . If that vector is initially zero, choose any direction when the radius first becomes positive; the resulting path is still continuous at the initial point. For a positive radius its direction lies in the path-connected sphere S^3 , so the direction may be changed to any chosen common direction.

Now join y to 0 by a line segment. The continuous function S is bounded on this compact segment. Choose $t = \epsilon > 0$ small enough that $\epsilon^4 S < 1/2$ throughout it. Keeping the common direction and varying the radius continuously gives a path along the whole segment. At $y = 0$, any two admissible positive values of t can be joined, again by radial rescaling. Thus all positive- t points lie in one path component. The sign symmetry proves the same for negative t . $\square$

Only the number of components is uniform. For example, if $H = 0$, each component has an S^3 factor in the real description. If the only residual is the constant 1, the corresponding radius closes at a finite positive t . We do not assert a single source-independent diffeomorphism type.

9 Why three weighted guards: a sharp restricted obstruction

The five-variable result is not obtained merely by deleting one guard from a sum of four identical ones. The weights and the use of the unit residual in the rational-point construction are essential.

9.1 Classification by total guard weight

Consider the more general architecture

$$F_{\mathbf{w}} = H + (tu - 1)^2 + \sum_{j=1}^k w_j(2z_j^2 - z_j), \quad w_j \in \mathbb{Z}_{>0}, \quad w_1 = 1, \quad (19)$$

and let $W = \sum_j w_j$. The assumption $w_1 = 1$ gives an especially simple uniform certificate at the prime two. Define

$$E = 4H, \quad D_j = w_j(4z_j - 1), \quad J_W = 8F_{\mathbf{w}} - 2E - \sum_j (4z_j - 1)D_j = 8R^2 - W.$$

A general elimination identity is

$$W(8 - W) = 4u(8R - W)(F_{\mathbf{w}})_u - (8R + 8 - W)J_W. \quad (20)$$

It follows by substituting $(F_{\mathbf{w}})_u = 2tR$ and $tu = R + 1$.

Theorem 9.1 (Universal smoothness in the weighted architecture). *Under $w_1 = 1$, the hypersurface (19) is smooth over $\mathbb{Z}$ for every quadratic source system if and only if*

$$W \in \{4, 16\}.$$

Proof. In the quotient by the equation and all its derivatives, $D_1 = 0$ gives $4z_1 = 1$, so two is invertible. Identity (20) says that $W(8 - W)$ is zero in that quotient. For $W = 4$ or $W = 16$ it is a nonzero signed power of two. The quotient is therefore the zero ring, proving that the Jacobian ideal is the unit ideal and the hypersurface is smooth.

For necessity, suppose an odd prime p divides $8 - W$. At the point $x = t = u = 0$ and $z_j = 1/4$ over $\mathbb{F}_p$, the equation and all its partial derivatives vanish. This works for every source, because H is homogeneous of degree four and has zero gradient at the origin. The case $W = 8$ gives the same counterexample for every odd prime.

If instead an odd prime p divides W , choose the zero source $H = 0$ and take $t = u = 1$, $x = 0$, and $z_j = 1/4$. This is again a singular point. Hence universal smoothness requires that both W and $|8 - W|$ be nonzero powers of two. Write $W = 2^a$. If $a < 3$, the odd factor of $8 - W$ is $2^{3-a} - 1$, which equals one only for $a = 2$. If $a > 3$, the odd factor of $W - 8$ is $2^{a-3} - 1$, which equals one only for $a = 4$. The case $a = 3$ was excluded. Thus $W = 4$ or 16 . $\square$

For identical guards $w_j = 1$, this says that exactly $k = 4$ and $k = 16$ give universal smoothness. In particular, simply using one, two, or three identical guards always introduces a bad odd prime: respectively seven, three, or five at the origin. The weighting $(1, 1, 2)$ keeps total weight four while using only three variables.

9.2 Why the still smaller weighting (1, 3) is insufficient

Two guards with weights (1, 3) also have total weight four, and therefore satisfy the smoothness theorem. They do *not*, however, give a dyadic point for every source. This is not merely a failure of our chosen point-construction algorithm.

Lemma 9.2 (A dyadic ternary-form obstruction). *If an integer N satisfies $N \equiv 10 \pmod{16}$, the equation*

$$X^2 + 3Y^2 + 2Z^2 = N$$

has no solution with $X, Y, Z \in \mathbb{Z}[1/2]$.

Proof. First, such a solution would have integral coordinates. Otherwise multiply by a smallest common denominator 2^d , $d \geq 1$, to obtain integers X_0, Y_0, Z_0 , not all even, with

$$X_0^2 + 3Y_0^2 + 2Z_0^2 = 2^{2d}N \equiv 0 \pmod{8}.$$

Reduction modulo four forces Z_0 even and X_0, Y_0 of the same parity. Primitivity then makes X_0, Y_0 odd, but the left side is four modulo eight, a contradiction.

For an integral solution, the even value of N forces X, Y to have the same parity. If both are odd, the left side is four or six modulo eight, not $N \equiv 2 \pmod{8}$. If both are even, write $X = 2a$, $Y = 2b$. Reduction modulo four makes Z odd, so $2Z^2 \equiv 2 \pmod{16}$. The equation would imply $a^2 + 3b^2 \equiv 2 \pmod{4}$, which is impossible. $\square$

Proposition 9.3 (Failure of the two-guard dyadic guarantee). *For the source consisting of three constant residuals equal to one,*

$$3t^4 + (tu - 1)^2 + g(z) + 3g(w) = 0 \tag{21}$$

has no point over $\mathbb{Z}[1/2]$, although its affine model is smooth over $\mathbb{Z}$.

Proof. A hypothetical dyadic point is a real point. Completing squares gives $3t^4 \leq 1/2$, while the same estimate as before gives $tu > 0$. Thus $0 < |t| < 1$. Write $t = a/2^k$ with a odd and $k \geq 1$. With $R = tu - 1$, the equation becomes

$$(2z - 1/2)^2 + 3(2w - 1/2)^2 + 2R^2 = 1 - 6t^4.$$

Multiplying all three coordinates on the left by 2^{2k} gives a dyadic representation by $X^2 + 3Y^2 + 2Z^2$ of the integer

$$N = 2^{4k} - 6a^4 \equiv 10 \pmod{16}.$$

This contradicts Lemma 9.2. Smoothness follows from total weight four and first weight one. $\square$

Corollary 9.4 (Five auxiliaries are optimal in the weight-four family). *Among finalizers of the form*

$$H + (tu - 1)^2 + \sum_j w_j g(z_j), \quad w_j > 0, \quad \sum_j w_j = 4,$$

which must be smooth over $\mathbb{Z}$ and have a dyadic point for every source, at least three guard variables are necessary. The choice (1, 1, 2) is sufficient, so five total auxiliary variables are optimal within this family.

[write] [In Corollary 9.4, $w_j \in \mathbb{Z}_{>0}$, as in (19); the display “ $w_j > 0$ ” is the source’s. The normalization $w_1 = 1$ of Theorem 9.1 is not assumed: the all-even weightings (4) and (2, 2) are excluded by the all-even argument of the proof below, and (1, 3) and (1, 1, 2) have a weight one after reordering.]

Proof. With one guard the only weight is four. With two guards the weights are, up to order, $(2, 2)$ or $(1, 3)$. All-even weights give a zero gradient modulo two for the zero source $H = 0$, and a nonempty singular fibre. The remaining pair $(1, 3)$ is excluded by Proposition 9.3. Theorem 3.1 proves sufficiency of three guards. $\square$

This is a restricted optimality theorem. It does not rule out a different unit gadget, a different nonnegative polynomial guard, a different total weight, or a construction with fewer variables that avoids this architecture entirely.

10 A concrete Turing-complete frontend

The finalizer does not depend on a particular language. Nevertheless, an executable frontend makes the semantics and the witness-count claims concrete. We use finite deterministic counter programs with natural registers and instructions of the following forms:

$$\text{inc}(j, q), \quad \text{dec}(j, q_+, q_0), \quad \text{halt}.$$

An increment adds one to counter j and jumps to q . A decrement instruction decrements a positive counter and jumps to q_+ , or leaves a zero counter unchanged and jumps to q_0 . There is one halt state, and it has an absorbing self-loop for padding.

10.1 Why this instruction set can support universal computation

For completeness, three counters suffice for a direct simulation of a Turing machine. Store the tape to the left and to the right of the current head in two stacks, keep the current head symbol in finite control, and use a third counter as scratch space. Encode a stack in a fixed base B using nonzero digits for tape symbols, with the empty stack encoded by zero. The outermost stack symbol is the least significant digit.

To push a digit d , replace X by $BX + d$. Repeatedly decrement X while adding B units to a scratch counter; then transfer the scratch counter back to X and add d . The fixed additions by B or d expand into finite sequences of increment instructions.

To pop, repeatedly decrement X , maintaining the remainder modulo B in finite control and incrementing the scratch counter after each complete block of B decrements. At the end, the scratch counter is the quotient and the control state holds the remainder. Transfer the quotient back to X . An empty stack returns the blank symbol. Scratch space is zero at the end of either operation.

A tape move pushes the updated head symbol onto one stack and pops from the other. Thus each transition expands into a finite counter program, with the unused stack preserved. This supplies an explicit simulation argument for universality of the program class. The small worked example below uses only one counter and is not itself a universal machine. Formalized reductions through Minsky machines and FRACTRAN provide an independent established context for this substrate choice [2]. [write] [Minsky showed that two counters already suffice for universal computation [11]; the three-counter simulation above is a direct argument, not a sharper count.]

10.2 A quadratic certificate for an external horizon

Fix a program with ℓ states, r counters, and a chosen horizon T . Expand instructions into edges e : increment edges, positive-decrement edges, zero-test edges, and the absorbing halt

edge. Let E be their number, p the number of positive-decrement edges, and g the number of all guarded edges (positive and zero). Write $s(e), t(e)$ for source and target states, and $\delta_{e,j} \in \{-1, 0, 1\}$ for counter updates.

Introduce natural variables

$$q_{s,i} \quad (0 \leq s \leq T, \ 0 \leq i < \ell), \quad c_{s,j} \quad (0 \leq s \leq T, \ 0 \leq j < r),$$

edge selectors $a_{s,e}$ for $s < T$, and one slack $d_{s,e}$ for every positive-decrement edge at every time $s < T$. The equations at time s are

$$\sum_e a_{s,e} = 1, \tag{22}$$

$$q_{s,i} = \sum_{e:s(e)=i} a_{s,e}, \quad q_{s+1,i} = \sum_{e:t(e)=i} a_{s,e}, \tag{23}$$

$$c_{s+1,j} = c_{s,j} + \sum_e \delta_{e,j} a_{s,e}. \tag{24}$$

For an edge testing counter j , impose

$$a_{s,e} c_{s,j} = 0 \quad \text{on a zero edge,} \tag{25}$$

$$a_{s,e} c_{s,j} - a_{s,e} - d_{s,e} = 0 \quad \text{on a positive edge.} \tag{26}$$

The initial one-hot state and initial counters are fixed by linear equations. Finally impose $q_{T,h} = 1$, where h is the halt state. Every displayed residual is of degree at most two.

Theorem 10.1 (Canonical bounded counter certificates). *The natural zero set of this quadratic system is empty if the program has not halted by time T , and is a singleton if it has. The same statement holds for its smooth quartic finalization, with five additional variables. The quadratic interface has*

$$n_T = (\ell + r)(T + 1) + (E + p)T, \tag{27}$$

$$m_T = (\ell + r) + T(1 + 2\ell + r + g) + 1. \tag{28}$$

Proof. Because the selectors are natural and sum to one, exactly one selector is one and all the others are zero. The state equations identify its source and target. No separate Boolean equation is needed. On a selected zero edge, (25) forces the counter to be zero. On a selected positive edge, (26) says $c_{s,j} = 1 + d_{s,e}$, so the counter is positive. On an inactive positive edge, that same equation forces its slack to be zero, rather than leaving an arbitrary witness.

Starting from the fixed initial state, exactly one operational edge is enabled. Induction on s therefore determines every selector, every counter, and every state variable uniquely. Active positive slacks equal the tested counter minus one; inactive slacks are zero. Conversely the actual padded run provides precisely this assignment and satisfies all transition equations. The endpoint equation holds exactly when the run has reached the absorbing halt state by time T .

There are $(\ell + r)(T + 1)$ state and counter variables, ET selectors, and pT slacks. The initial rows number $\ell + r$. Each step has one selector-sum row, 2ℓ state rows, r update rows, and g guard rows. There is one final endpoint row. This gives (27)–(28), and Theorem 3.1 gives the last assertion. $\square$

The proof depends on natural coordinates. Over unrestricted integers a sum of selectors equal to one does not make them one-hot. The integer zero-set theorem for the finalizer still holds, but it refers to the integer zeros of the source equations, which need not describe legal

counter executions. A separate domain encoding is required before using this frontend as an integer execution certificate.

[write] [Theorem 10.1 is a second presentation of the counter-program adapter of *canonical-diophantine-certificates*: its one-hot control equations, the zero guard (25), which is that report’s equation `cdc:eq:zero-guard`, the externally fixed horizon of its Section `cdc:pt:sec:boundary` (arity growing with T), and the halt self-loop caveat of its Section `cdc:ct:sec:substrates`. New here are only the exact counts (27)–(28) and the positive-edge slack equation (26), with natural selectors summing to one in place of Boolean rows. The worked example below is printed as an illustration of the finalizer.]

10.3 Worked example: a countdown program

Take two states: state zero executes `dec(0,0,1)`, and state one halts. The three edges are D (decrement and loop), Z (zero-test and enter halt), and H (halt self-loop). For input counter value two, the run is

$$(0, 2) \xrightarrow{D} (0, 1) \xrightarrow{D} (0, 0) \xrightarrow{Z} (1, 0) \xrightarrow{H} (1, 0) \longrightarrow \dots$$

Thus horizon two rejects; horizon three accepts; larger horizons add only forced halt padding. Here

$$\ell = 2, \quad r = 1, \quad E = 3, \quad p = 1, \quad g = 2, \quad n_T = 7T + 3, \quad m_T = 8T + 4.$$

For $T = 3$ the source has 24 variables and 28 residuals. The exported smooth equation has 29 variables, exact degree four, and 97 nonzero monomials. Its natural zero set is a singleton.

The exported rational point deliberately need not coincide with the computational witness. The source constant statistic is $c = 9$; the reference constructor takes $K = 4$ and uses

$$238 = 0^2 + 2^2 + 3^2 + 15^2 = 3^2 + 15^2 + 2 \cdot 1^2 + 2 \cdot 1^2.$$

All 24 source coordinates are set to zero in that rational point, with

$$t = \frac{1}{4}, \quad u = \frac{17}{4}, \quad (z_1, z_2, z_3) = \left(\frac{11}{32}, \frac{23}{32}, \frac{9}{32} \right).$$

Here $H = 9/256$, $R^2 = 1/256$, and the weighted guard sum is $-10/256$. The endpoint condition is not being claimed to hold by itself at this rational tuple. Only the full finalized equation vanishes.

11 Consequences for unbounded computation

11.1 An undecidable smooth quartic class

The MRDP theorem identifies computably enumerable relations with Diophantine relations. The existence of a fully mechanized proof, including both directions and machine translations, is established in the work of Larchey-Wendling and Forster [2]. Here MRDP is a background theorem, not a result reproved by the finite counter compiler.

Theorem 11.1 (A geometrically restricted complete problem). *Integer solvability is computably enumerable complete, under computable many-one reductions, even on the explicit class of quartic polynomials produced by (3). Every instance in this class comes with*

- (a) *a smooth affine model over $\mathbb{Z}$ with geometrically integral fibres;*
- (b) *a bounded-degree integral Jacobian certificate;*
- (c) *a constructible nonnegative dyadic rational point;*
- (d) *integral points over every $\mathbb{Z}_p$ and solutions modulo every integer;*
- (e) *a Zariski-dense set of rational points and exactly two real path components.*

The analogous statement for natural solvability holds on the natural quadratic-lifting version of the class.

Proof. Given an integer polynomial equation, apply the integer version of Lemma 2.2, then the finalizer. Integer solvability is preserved at both stages. Equivalently, to start from a natural MRDP representation, introduce integer variables for each original natural witness and impose that it is a sum of four integer squares. These additional equations are quadratic, and circuit lifting deals with the remaining polynomial equations. This second route does not preserve witness uniqueness at the four-square stage, and no such assertion is needed for undecidability.

The known computably enumerable completeness of Diophantine solvability therefore transfers to the displayed subclass. Membership remains computably enumerable by enumerating integer tuples and evaluating the polynomial. All geometric and arithmetic promises follow from Theorem 3.1; their certificates can be generated with the reduction and do not require discovering an integer solution.

For natural solvability, use the natural circuit lifting in Lemma 2.2. The finalizer then preserves the full natural zero set bijectively. $\square$

This theorem is compatible with the older smooth-variety undecidability result cited in the introduction. Its content is a specific simultaneous normal form, including an everywhere smooth *integer* model and explicit positive evidence for all the easier existence notions.

[write] [The computability core of Theorem 11.1, MRDP followed by unique circuit lifting and a sum of squares of degree four, is the quartic normalization of *canonical-diophantine-certificates*, Proposition `cdc:of:prop:quartic`; the new content is the package (a)–(e). In the formal project beside the collection, MRDP is the Lean theorem `Diophantine.mrdp` (`Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean`), and the natural-to-integer four-square substitution of the second route appears as `exists_sqTerm_eq` in `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1978/Enumeration.lean`; neither formalizes any statement of this report.]

11.2 Fixed arity versus external duration

Fixing a universal Diophantine source relation gives a fixed finite quadratic circuit and therefore a fixed number of variables in the finalized family. This is a legitimate fixed-arity unbounded statement, because the source already comes from MRDP or another proved unbounded representation.

By contrast, the counter frontend of Section 10 creates n_T variables and m_T equations. Its horizon is an external compiler argument. The facts that each accepting bounded run has one witness and that finalization adds only five variables do not combine to give a fixed-arity unique representation of arbitrary halting.

More explicitly, even the disjoint union of the bounded accepting certificates has a copy for every T after the first halting time because of padding. Restricting to a first-halting horizon

would remove that particular repetition, but compiling a variable-length history into one fixed polynomial while preserving all multiplicities is an additional problem. It is not accomplished by treating a loop in the compiler as a polynomial variable.

11.3 Input parameters and total degree

Suppose a master family uses an input parameter a only through a loader $v - a = 0$, while the remaining quadratic equations have fixed integer coefficients. If a is treated as a coefficient during homogenization, the loader becomes

$$tv - at^2.$$

Its square has total degree six when a is counted as a coordinate, although it is quartic in witness variables for every fixed a . Therefore this straightforward master presentation has

$$\deg_{\text{all coordinates}} \mathcal{F} \leq 6, \quad \deg_{\text{witnesses}} \mathcal{F}(a; \cdot) = 4.$$

It is smooth *relative to the input base*: the same Jacobian identity uses only witness derivatives and continues to hold with a in the coefficient ring.

Alternatively, homogenizing a as a source variable keeps the total polynomial quartic and gives a smooth total space. It does not by itself prove that every fixed- a fibre is smooth, because the Euler identity then uses the derivative in the a direction. These two degree/smoothness contracts must not be conflated.

11.4 A separation between rational and integer search

Corollary 11.2 (No computable bound for the least integer witness). *There is no computable function of the finite source description that bounds the height of an integer point on every solvable compiled instance. This remains true even when the dyadic point and Jacobian certificate are included as additional input data.*

Proof. Such a function would turn integer solvability into a finite search over a box of its computed radius, deciding the complete problem in Theorem 11.1. The additional data are computably produced from the same source description and do not change the reduction. $\square$

The rational side is starkly different: a point of polynomially bounded coordinate height in $\sqrt{c+1}$ is always available. Thus small rational solutions do not bound the size of the first integer solution. The exact height transfer in Corollary 4.2 also shows that this separation is not caused by huge integer auxiliary values introduced by the finalizer.

12 A direct higher-degree extension

Quadratic lifting is useful for degree four, but it is not needed for smoothness itself. A power-of-two padding gives a second tradeoff: keep only five new variables and permit a larger degree.

Theorem 12.1 (Power-of-two homogeneous padding). *Let $f_i \in \mathbb{Z}[x]$ have degrees at most D . Choose a positive power of two $d \geq \max\{1, D\}$ and set*

$$q_i(x, t) = t^d f_i(x/t), \quad H = \sum_i q_i^2, \quad F = H + (tu - 1)^2 + g(z_1) + g(z_2) + 2g(z_3),$$

where homogenization is interpreted polynomially. Then the natural and integer zero-set correspondences of Theorem 3.1 still hold. The affine model is smooth over $\mathbb{Z}$ with geometrically integral fibres, and it has dyadic points and integral points at all primes. Its degree is at most $\max\{4, 2d\}$.

Proof. Nonnegativity, the unit equation, and $q_i(\varepsilon a, \varepsilon) = \varepsilon^d f_i(a)$ prove the same lattice correspondences. The rank-three argument in the guard variables proves geometric integrality, and the characteristic-two fibre is again a polynomial graph.

For smoothness, work in the quotient ring by F and all its derivatives. The equation $D_1 = 4z_1 - 1 = 0$ makes two invertible. The other guard coefficients are one and two, hence are also units in this quotient, and all $z_j = 1/4$. Euler's identity now gives

$$\sum_i x_i F_{x_i} + t F_t - u F_u = 2dH = 0.$$

Since d is a power of two, $H = 0$. The equation $F = 0$ therefore gives $R^2 = 1/2$. The derivative $F_u = 2tR = 0$, multiplied by u , gives $R(R + 1) = 0$, so $R = -1/2$. Squaring contradicts $R^2 = 1/2$ by forcing $1/4 = 0$. Since four is a unit, the quotient is the zero ring. This is exactly the unit-Jacobian-ideal criterion for smoothness.

For dyadic points choose a power of two K with $K^{2d} > 2c$ and represent

$$K^{2d} - 2c = A_1^2 + A_2^2 + 2A_3^2 + 2B^2.$$

Take $x = 0$, $t = 1/K$, $R = B/K^d$, $u = K + B/K^{d-1}$, and $z_j = 1/4 + A_j/(2K^d)$. The same substitution as before proves $F = 0$. This point is integral at odd primes; at two the equation $c + 2z_1^2 - z_1 = 0$ again gives a lift with $t = u = 1$. $\square$

Choosing the smallest such d gives $d < 2D$ for $D \geq 1$, except that $d = D$ when D already is a power of two. Thus there is a simple variable/degree tradeoff between direct padding and unique quadratic circuit lifting. Only the quadratic version is implemented in the supplied reference compiler; this section is a proved extension, not an additional software feature.

13 Implementation, exact checks, and reproducibility

13.1 Files and interfaces

The archive contains a self-contained L^AT_EX source and compiled PDF, three Python modules, two expanded polynomial exports, and an execution report. The main software files are

File	Purpose
<code>code/smooth_compiler.py</code>	Validated quadratic input, homogeneous lift, quartic export, Jacobian certificate, dyadic point, and compatible 2-adic residues.
<code>code/counter_frontend.py</code>	Counter-program semantics and the bounded quadratic frontend.
<code>code/verify.py</code>	Exact symbolic identities, independent sparse polynomial evaluation, finite exhaustive checks, and export generation.

[write] [Shipped names. In this report’s directory the delivered files carry the prefix `13-smooth-quartic-`: `code/smooth_compiler.py`, `code/counter_frontend.py`, `code/verify.py` and the `Makefile` are `code/13-smooth-quartic-*`; the two exports `examples/countdown_T3.json` and `examples/inconsistent.json`, the records `verification/results.json` and `verification/document_qa.json`, and `requirements.txt` are `data/13-smooth-quartic-*`; `RESEARCH_STATUS.md` and `SOURCE_AUDIT.md` are at the report root under the same prefix. The delivered PDF, the run log (a copy of `results.json`) and the checksum file are not shipped; the PDF of this report is a new build. Delivered names below are the source’s. The programs import each other by their delivered names and write into `examples/` and `verification/` beside `code/`, so they must be run in a copy with the delivered layout; the report’s README gives the recipe.]

A `QuadraticSystem` copies its variables and residuals into immutable tuples. It rejects undeclared symbols, repeated variable names, nonintegral coefficients, floating-point coefficients, and residual degree above two. The finalizer checks auxiliary-name collisions. Counter-program validation checks register ranges, state targets, instruction fields, nonnegative initial counters, and an exact nonnegative integer horizon.

The dyadic constructor computes a four-square decomposition by a finite pair table and converts it to weights $(1, 1, 2, 2)$ using an equal-parity pair. The pair-table routine has a declared default input cap of one million to prevent an unexpected large allocation. That is a software resource limit, not a mathematical exception. The caller may supply a four-square representation for independent exact checking instead.

13.2 Recorded checks

The included run used Python 3.13.5 and SymPy 1.14.0 with seed 20261002. All 21,128 recorded assertions passed. The principal groups are shown in Table 2; the exact full breakdown is in `verification/results.json`. [write] [Shipped as `data/13-smooth-quartic-results.json`. A rerun at placement (Python 3.14.4 and 3.13.5, SymPy 1.14.0, on a copy) and the Hilbert’s-tenth review [9] both reproduced all 21,128 assertions and the 40 group counts.]

Check group	Assertions
Integer tuples in three explicit six-dimensional boxes	10,125
Natural subboxes of those same boxes	648
One-step countdown assignments, exhaustively compared to semantics	2,048
Ternary obstruction, all triples modulo sixteen	4,096
Primitive triples modulo eight in the same obstruction	448
Independent sparse-export evaluations on seeded tuples	520
Odd-prime reductions of the constructed dyadic points	260
Four-square and weighted-four-square replay for $0 \leq N \leq 200$	402
Four nested Jacobian/Euler identities on 27 source systems	108
Expanded Bézout identities and coefficient-degree bounds	54
Malformed-input rejection checks	21

Table 2: Selected groups of exact checks; not a replacement for the proofs.

The source systems used for symbolic checks include zero-variable cases, a zero system, constant inconsistent systems, simple quadratic equations, 20 seeded two-variable systems, and the countdown certificate. The integer-box test ranges over $x, t, u \in \{-2, -1, 0, 1, 2\}$ and each of the three guard variables in $\{-1, 0, 1\}$. There are $5^3 3^3 = 3375$ tuples per source. These ranges are recorded so that a finite test is not mistaken for an unbounded exhaustive verification.

For the counter frontend, 140 initial-input/horizon combinations are checked across a one-counter and a two-counter program. The transitions, endpoint, variable count, residual count, and nonnegativity checks are separate assertions. The one-step exhaustive check independently compares the polynomial rows with the operational transition relation, rather than merely checking the one run returned by the compiler.

The sparse JSON evaluator multiplies integer coefficients by powers of supplied coordinates; it does not call SymPy substitution into the original residual list. On seeded tuples its result is compared with direct homogeneous-residual evaluation. For the Jacobian identity, both the nested formula (8) and the expanded coefficient formula (10) are checked symbolically from the actual emitted F .

13.3 What the checks establish and what they do not

The symbolic identities certify the algebraic calculations for the exported examples and sampled systems. The paper supplies the universal derivations. The finite checks can reveal implementation errors in lifting, signs, weights, inactive slacks, variable order, and exported coefficients. They do not prove the general source semantics or replace the scheme-theoretic arguments about smoothness and geometric integrality.

No Lean or Coq process was run for this manuscript. No existing repository axiom audit was rerun. The ordinary proofs of rational density, the number of real components, the higher-degree padding theorem, and the general undecidability reduction are not advertised as consequences of a Python unit test.

To reproduce the package from its root directory:

```
python -m pip install -r requirements.txt
python code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A minimal use of the finalizer is

```
import sys
sys.path.insert(0, "code")
import sympy as sp
from smooth_compiler import QuadraticSystem, compile_smooth
from smooth_compiler import dyadic_point

x, y = sp.symbols("x y")
source = QuadraticSystem((x, y), (x - y*y, y - 2))
certificate = compile_smooth(source)
integer_point = certificate.lift((4, 2))
assert certificate.evaluate(integer_point) == 0
rational_point, metadata = dyadic_point(certificate)
assert certificate.evaluate(rational_point) == 0
assert certificate.jacobian_certificate()["unit"] == 1
```

[write] [Both listings assume the delivered layout; in this report's directory neither `requirements.txt` nor `code/verify.py` exists under that name, and `verify.py` rewrites the two exports and `results.json` in place. Use the copy recipe of the report's README; on this machine the interpreter is `py`.]

14 Formalization plan and theorem boundaries

A useful formalization should mirror the actual proof boundary rather than importing the final statement as a large opaque axiom. The algebraic core can be separated from the geometry.

First formalize integer guard nonnegativity and its unique zero, the two units of $\mathbb{Z}$, and homogeneous lifting. These yield the natural bijection, the signed integer bijection, and the height correspondence. Their proofs need no algebraic geometry.

Next formalize the identities (5), (6), and (10) over a general commutative coefficient ring. One may represent the certificate as a finite tuple of polynomials and check the single ring identity. An exported, independently checked identity is a natural target, but a theorem connecting such a checker to polynomial semantics is needed before it becomes a trusted bridge.

The geometric layer should separately connect a unit Jacobian ideal to relative smoothness and prove the rank-three irreducibility argument. The characteristic-two polynomial-graph isomorphism is elementary and should not be subsumed under a characteristic-zero smoothness theorem. The rational/local layer then needs four-square existence, the parity conversion, exact rational substitution, and compatible 2-adic lifting.

Finally, the counter frontend should prove its natural one-hot lemma, per-edge correctness including inactive slacks, and induction on the externally fixed horizon. Its output can then feed the generic finalizer without repeating the geometric proof for every language.

Several boundaries are structural, not minor missing optimizations:

- A smooth affine model is not a smooth proper model. The points at infinity of a homogenized projective closure are outside the main theorem.
- Dyadic solubility of this class does not decide arbitrary equations over $\mathbb{Z}[1/2]$ or $\mathbb{Q}$. The reduction preserves integer or natural solvability, not rational solvability.
- Witness preservation does not establish single-fold or finite-fold MRDP. It preserves whatever multiplicities the source already has.
- External horizon bounds are not compressed by adding five variables. Fixed arity is obtained only from an already fixed-arity source relation.
- The sharp five-auxiliary lower bound proved here is restricted to the positive weight-four guard architecture; it is not a general lower bound for smooth Diophantine representations.

15 Further research questions and concrete next targets

The following questions are proposals arising from this construction. Unless explicitly stated otherwise, they are not assertions that no answer exists in the literature. A dedicated priority search should accompany any attempt to publish a claimed resolution.

Research question 15.1 (Four auxiliary variables with the full package). Can another integer-polynomial finalizer preserve every natural zero set bijectively, remain quartic and smooth over $\mathbb{Z}$, and guarantee a dyadic point for every source while adding at most four variables?

Proposition 9.3 excludes the most immediate candidate, not all candidates. A successful construction must avoid the ternary-form obstruction rather than merely find a better algorithm for that form. Promising search spaces include unequal guard centres, nonquadratic guards whose total degree stays four, or a replacement for the product-unit constraint.

Research question 15.2 (Other total weights). Within (19), can two positive guards of total weight sixteen give the complete dyadic guarantee for every quadratic source?

The smoothness classification allows total weight sixteen. It leaves a finite family of two-guard choices with first weight one, or a slightly larger family after replacing that normalization by a coprime-weight condition. The rational point problem becomes a ternary quadratic-form problem coupled to fourth-power scaling. An exact local obstruction or a uniform construction would materially extend the sharp result above.

Research question 15.3 (A totally quartic, relatively smooth universal family). Can one obtain a single polynomial of total degree four in both input parameters and witnesses, with every integer-input fibre smooth over $\mathbb{Z}$, while preserving a universal natural relation and retaining the same local-solubility guarantees?

Section 11.3 gives a total-degree-six master family and explains why homogenizing the parameter changes the relative-smoothness claim. An input loader compatible with both total degree and the relative Euler identity is the specific missing component.

Research question 15.4 (A smaller Jacobian certificate). Can the degree-seven expanded identity be replaced uniformly by a certificate whose products all have degree at most six, or by a shorter arithmetic circuit with the same universal coefficient contract?

This is a certificate-optimization problem, not an attempt to improve a universal halting polynomial's operation count by ignoring its frontend. The precise optimization object should include the equation, all derivative costs, and the coefficient tuple in (9).

Research question 15.5 (Controlled higher topology). Which topological invariants of the two real components can be read from the real zero set of the source system, and can the construction be modified to make every real component contractible?

The model (18) gives a concrete starting point. Its fibres change from a capped sphere family to a cylinder at source real zeros. The number of components is fixed, but a full homotopy or diffeomorphism classification is not supplied here.

Research question 15.6 (Strong approximation and integral obstructions). What can be said about strong approximation, and about more refined obstructions to integral points, on these explicit affine varieties?

All local integral points exist and rational points are dense in the real locus, but these statements do not imply simultaneous approximation at many places with integral control away from them. A meaningful answer must respect the undecidability theorem: no proposed effectively complete, uniform terminating obstruction procedure can decide all compiled inputs.

Research question 15.7 (Density of restricted-denominator points). For which source systems are the dyadic points dense in a real component? Can a bounded set of additional denominator primes guarantee density?

The proof of Theorem 8.1 uses rational parametrizations with unrestricted denominators. Replacing them by dyadic parameters does not preserve all denominator conditions automatically. This is a distinct arithmetic approximation problem, not an immediate corollary of the existence construction.

Research question 15.8 (Smooth models over other arithmetic rings). Which discretely ordered domains or rings of S -integers admit analogues with exact control of unit-induced witness multiplicity?

The integer proof uses both the positivity gap of $2z^2 - z$ on the lattice and the fact that the unit group is $\{\pm 1\}$. Neither property carries over unchanged to a general ring of integers. A useful extension should state exactly what replaces the lattice order and how the unit group acts on fibres.

Research question 15.9 (A verified finalizer integrated with repository frontends). Can the five-variable transformation be formalized once and instantiated on the repository’s existing causal, queue, heap, and interaction-net interfaces without changing their semantic contracts?

A practical first milestone is the homogeneous lift plus the explicit Jacobian certificate. A second is a proved checker for the sparse JSON format. A third is one full frontend-to-smooth-quartic theorem with an explicit distinction between natural and integer witnesses. [write] [Open at the write. The interfaces named are Parts of the collection’s report *canonical-diophantine-certificates*, which are research text, not formal developments; the Lean project `Computability/HilbertTenthProblem` has no smoothness or finalizer module, and the review’s portable helper [9] is an independent replay, not a proved checker.]

Research question 15.10 (A detailed publication and priority audit). Which parts of the combined normal form are already implicit in known smoothness reductions or in work on integral local-to-global phenomena?

The introduction identifies a definite antecedent and avoids presenting the general smooth-undecidability phenomenon as new. A systematic comparison should separately track good reduction at every prime, fixed auxiliary count, dyadic sections after specialization, the exact natural bijection, and the bounded-degree Jacobian certificate. These properties should not be collapsed into the vague phrase “smooth Diophantine representation.” [write] [Open at the write. The Hilbert’s-tenth review [9] checked Poonen’s Lemma 4.1 and Corollary 4.3, the mechanized MRDP record and the Stacks Project criterion, and states that it was not a complete historical priority search.]

16 Conclusion

The construction separates three layers that are easily confused. The source quadratic equations contain the computational semantics. The integer gap of the guards and the unit equation preserve their witnesses. The homogeneous scaling and weighted guards alter the surrounding arithmetic geometry so that every geometric fibre is smooth and integral, all integral local points exist, and rational points are plentiful.

The resulting polynomial is

$$\sum_i q_i(x, t)^2 + (tu - 1)^2 + (2z_1^2 - z_1) + (2z_2^2 - z_2) + 2(2z_3^2 - z_3).$$

It has exactly five additional variables. Its geometric regularity is certified by an integral identity of bounded degree, not by a generic choice, numerical evidence, or an unbounded singularity search. Its natural witnesses are unchanged; its integer witnesses form exactly two sign-related copies. The failed two-guard alternative shows that the arithmetic point construction imposes a genuine additional constraint beyond smoothness.

This does not settle a classical undecidability conjecture or erase the finite-fold boundary. It provides a reusable explicit normal form: even a smooth affine quartic with everywhere good affine reduction, small dyadic points, and only two real components can carry the full difficulty of integer computation.

A The complete countdown residual list

For the input two and horizon $T = 3$, use variables

$$q_{s,0}, q_{s,1}, c_s \quad (0 \leq s \leq 3), \quad a_{s,D}, a_{s,Z}, a_{s,H}, d_s \quad (0 \leq s < 3).$$

There are $3 \cdot 4 + 4 \cdot 3 = 24$ of them. The initial rows are

$$q_{0,0} - 1, \quad q_{0,1}, \quad c_0 - 2.$$

At each $s = 0, 1, 2$, include the following eight residuals:

$$\begin{aligned} & a_{s,D} + a_{s,Z} + a_{s,H} - 1, \\ & q_{s,0} - a_{s,D} - a_{s,Z}, \quad q_{s,1} - a_{s,H}, \\ & q_{s+1,0} - a_{s,D}, \quad q_{s+1,1} - a_{s,Z} - a_{s,H}, \\ & c_{s+1} - c_s + a_{s,D}, \\ & a_{s,D}c_s - a_{s,D} - d_s, \quad a_{s,Z}c_s. \end{aligned}$$

The final row is $q_{3,1} - 1$. Thus the total number is $3 + 3 \cdot 8 + 1 = 28$. The source witness is

$$\begin{aligned} (q_{0,0}, q_{0,1}) &= (q_{1,0}, q_{1,1}) = (q_{2,0}, q_{2,1}) = (1, 0), \\ (q_{3,0}, q_{3,1}) &= (0, 1), \quad (c_0, c_1, c_2, c_3) = (2, 1, 0, 0), \\ (a_{0,D}, a_{0,Z}, a_{0,H}) &= (a_{1,D}, a_{1,Z}, a_{1,H}) = (1, 0, 0), \\ (a_{2,D}, a_{2,Z}, a_{2,H}) &= (0, 1, 0), \quad (d_0, d_1, d_2) = (1, 0, 0). \end{aligned}$$

Add $(t, u, z_1, z_2, z_3) = (1, 1, 0, 0, 0)$ to obtain the unique natural zero of the finalized equation.

For an entirely explicit compact polynomial, apply the following rule to each row above: multiply its quadratic terms by 1, its linear terms by t , and its constants by t^2 . Square the 28 resulting homogeneous polynomials, add them, and then add $(tu - 1)^2 + g(z_1) + g(z_2) + 2g(z_3)$. This is a complete expression, not an appeal to an unspecified universal representation. Its expanded 97-term coefficient list is in `examples/countdown_T3.json`.

B The export and verification boundary

Each JSON export records an ordered variable list and a list of terms. A term has an integer coefficient and an exponent vector of the same length as the variable list. Its mathematical meaning is

$$c \prod_{j=1}^N v_j^{e_j}.$$

Zero coefficients are omitted; the degree is the maximum sum of exponents among retained terms. The format also records the source residuals for human inspection, the guard weights, a rational point as exact strings, the four-square conversion data, and compatible 2-adic residues.

The equation is reconstructed for independent evaluation from the coefficient/exponent data. The source residual strings are not interpreted as trusted assertions about machine semantics. A future formal checker should validate the term format, coefficient ring, variable order, and identity semantics before using such an export as a theorem input.

The file `examples/inconsistent.json` uses the constant source residual 1 with one unused source variable. It therefore has one more free source coordinate than Example 7.4; it is the

product of that example with an affine line. The printed five-variable equation and its rational point are checked separately by the zero-variable test. This distinction is intentional and is not a mismatch in the auxiliary variable count.

[write] [The two exports are shipped as `data/13-smooth-quartic-countdown_T3.json` and `data/13-smooth-quartic-inconsistent.json`.]

C Provenance and scope of the source review

The source comparison consisted of the repository root overview, the Hilbert-tenth-problem overview, the canonical-certificate collection README, and a pinned canonical-certificate research-status file. The last file was read at commit

`a845ab5d0f4591294ef3c2d92feaeb5825a8d543`

and had Git blob identifier

`e768b2284c6723d21d77edfb24f786ccdb83585a`.

The source review did not establish the absence of an equivalent formula elsewhere in the repository or literature.

The external mathematical references used directly are Poonen’s smoothness reduction, the primary account of mechanized MRDP by Larchey-Wendling and Forster, and the Stacks Project’s smoothness criterion. The general four-square theorem is used as classical background; its application to weighted $(1, 1, 2, 2)$ representation is proved in the text. The explicit five-variable formula, the universal Bézout identity, the dyadic construction, the weighted two-guard obstruction, and the detailed counter frontend were developed in this manuscript and checked as reported. Their status is that of proved research claims awaiting independent mathematical review, not established publication priority.

D Provenance of this report

[write] The report was built from *one* manuscript, batch-79 manuscript 13 of ProveIt’s incoming reports (cluster J1 of that batch). It arrived as the archive `Smooth_Quartic_Diophantine_Certificates.zip` in commit `ef2fc7990` (2 October 2026), with inner directory `smooth_diophantine/`: the main file `article.tex` (1,733 lines, printed here), a 28-page PDF, a delivery README, research-status and source-audit notes, three Python modules, a Makefile, two exports, two verification records, a run log, a requirements file and a checksum file (16 members; the checksum file verified 15 of 15 at placement). Commit `224ca41df` placed it as this new report and staged its programs, data and notes byte-identical to the delivery.

Pin. The manuscript names the repository commit `a845ab5d0f4591294ef3c2d92feaeb5825a8d543` and the blob `e768b2284c6723d21d77edfb24f786ccdb83585a` of `canonical-diophantine-certificates/01-causal-traces-RESEARCH_STATUS.md` (Section 1.1 and Appendix C); its source audit also names the root README blob `bc4c514b61e3`. Both blobs are unchanged at the write.

Contribution. Everything in the report except the text marked [write]: the finalizer and Theorem 3.1 with its proofs, the size bounds, the weighted-guard classification and the restricted

optimality, the counter frontend and its worked example, the undecidability corollaries, the power-of-two padding, the implementation and its recorded checks, ten research questions, and two appendices on the countdown certificate and the export boundary.

Where the write had to choose.

- *Placement.* The manuscript post-processes any quadratic system, CDC’s or not, and its subject is the arithmetic geometry of the final equation rather than a substrate, so it was placed as a new report rather than as a further Part of *canonical-diophantine-certificates* after that report’s Part XI (the alternative recorded at placement).
- *Re-proved results.* Lemma 2.2, Theorem 10.1 and the computability core of Theorem 11.1 re-present results of that report. They are kept in full, because later proofs use them, with [write] pointers naming the host labels and no novelty claimed.
- *Numbering.* The editorial preface is unnumbered, so the manuscript’s section, statement, equation and table numbers are unchanged; this appendix is Appendix D, after the manuscript’s Appendices A–C.
- *Text kept as delivered.* The title page, the author line (“Prepared for Vladimir Reshetnikov / Developed with ChatGPT”) and the PDF metadata are the source’s; one [write] sentence was added to the status box. The display “ $w_j > 0$ ” of Corollary 9.4 is kept, with a [write] note giving $w_j \in \mathbb{Z}_{>0}$. No symbol was renamed and no normalization changed; the preface’s notation table lists the reused letters.
- *Added text.* The preface; [write] notes in Sections 1.1, 1.2, 2 (after Lemma 2.2), 7 (after Lemma 7.1), 9 (after Corollary 9.4), 10, 11, 13 (shipped file names, reruns) and 15 (Questions 15.9 and 15.10) and in Appendix B; this appendix; and the bibliography entries [7, 8, 9, 10, 11].

Files. The report directory holds this text (`article.tex`, `article.pdf`), a report README, the delivered `RESEARCH_STATUS.md` and `SOURCE_AUDIT.md` as `13-smooth-quartic-RESEARCH_STATUS.md` and `13-smooth-quartic-SOURCE_AUDIT.md`, the programs and Makefile in `code/` and the exports, records and requirements file in `data/`, all under the prefix `13-smooth-quartic-`. Not shipped: the delivered PDF and README, the run log (the results record plus one newline) and the checksum file. They survive in the archive of commit `ef2fc7990`.

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